

Pratham Kwatra

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WORK EXPERIENCE

Susquehanna International Group (SIG)

Philadelphia, PA

- *Software Engineer, Full-time*

July 2024 - Present

- **Eliminated critical race conditions** in the quote book logic by engineering a custom **TCP socket server** to emulate the NASDAQ SQF protocol, implementing manual packet framing and buffer management.
- **Achieved lock-free concurrency** by replacing 4,000+ dictionaries with a transaction queue architecture, removing context-switching overhead and ensuring data consistency for high-frequency updates.
- Enabled large-scale regression testing of **2,000+ live trading instances** by designing a "Safety Mode" system that audits full runtime configurations and captures state snapshots immediately prior to market connection.
- Automated the build and deployment pipeline using **TeamCity** and **Octopus Deploy**, integrating the emulator into the daily testing suite to reduce manual QA cycle time by 80%.

- *Software Engineer Intern, Monitoring Management*

March 2023 - September 2023

- Engineered an end-to-end monitoring and alerting solution (**ASP.NET Core, Typescript**) to secure and ensure uptime for 6,000+ critical production servers.
- Designed a high-throughput distributed service to ingest and process over **1.25 million log entries per minute**, streaming data into a partitioned **MS SQL Server** for anomaly detection.
- **Reduced deadlocks by >90%** and tripled throughput by diagnosing Entity Framework's internal **MERGE** lock contention and refactoring the persistence layer to bypass unnecessary read-verifications.
- Implemented a **server-based sharding strategy** for log ingestion, ensuring balanced load distribution across database partitions and preventing write-heavy hotspots during market open.

- *Software Engineer Intern, Middleware*

March 2022 - September 2022

- Mitigated resource leaks and security vulnerabilities by developing a **Python** and **Robot Framework** monitoring system to detect and alert on orphaned processes.
- Restored high availability for core trading services by analyzing and resolving a persistent deadlock in a proprietary, low-latency **C++** communication system.
- Automated the analysis of **process memory dumps** to proactively identify memory leaks in long-running **C++** trading processes before they caused critical OOM crashes.

Drexel University - College of Computing and Informatics

Philadelphia, PA

- *Backend Engineer, Capstone Project*

September 2023 - June 2024

- Reverse-engineered Stanford's MOSS plagiarism detection algorithm, analyzing its string-matching and statistical techniques to identify vulnerabilities and inform development of a new tool.
- Engineered a distributed data ingestion pipeline using **Traefik** and **FastAPI** to preprocess and queue massive volumes of source code submissions for algorithmic analysis.
- Deployed the analysis engine via **Docker containers**, ensuring consistent execution environments across distributed compute nodes and enabling horizontal scaling.

EDUCATION

Drexel University

June 2024

Bachelor of Science in Computer Science

GPA: 3.87/4.0

Honors: Magna Cum Laude

Concentrations: Systems and High-Performance Computing, Computing Systems

Relevant Coursework:

- Probability & Statistics, Linear Algebra, Database Systems, Algorithms, Data Structures
- Cloud Software Engineering (G), IT Security, Computer Forensics, Operating Systems, Computer Networks, Concurrent Programming